SeqFedEDT: Accelerating Sequential Federated Learning on non-IID Data via Element-Wise Decoupled Training

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***Abstract*—Sequential federated learning (SFL) trains models collaboratively across clients in a chain manner. This order shows communication efficiency compared to traditional FL in a parallel manner with a star topology. However, SFL can fail to produce stable training results when clients have significant statistical heterogeneity among their local data distributions. To address these challenges, we propose a novel element-wise model decoupling framework named `SeqFedEDT` that accelerates SFL training by separating model parameters of each client into a shared subset for global knowledge collaboration and a personalized subset for migrating data heterogeneity. We explore three types of parameter contribution scoring metrics based on gradient, Fisher information, and parameter importance (PI) for personalized parameter selection. In addition, we propose a quantile-based thresholding mechanism to separate shared and personalized subsets and explore the best performance quantile selection in numerical studies. Extensive experiments demonstrate that `SeqFedEDT` outperforms eight state-of-the-art methods across diverse datasets and heterogeneity scenarios.**

***Index Terms*—Sequential federated learning, element-wise model decoupling, parameter importance scoring, heterogeneous data distributions, personalized federated learning.**

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All code and results are available at https://github.com/tian0920/SeqFedEDT.

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I. Introduction

SEQUENTIAL Federated Learning (SFL) is a representative FL paradigm that trains models sequentially across clients while preserving the decentralized nature of their data [1], [2], [3], [4], [5], [6], [7], [8]. Unlike traditional parallel FL (PFL) with star-topology training (e.g., `FedAvg` [9], [10] with periodic all-gather aggregation synchronization), SFL employs chain ordering cooperation for communication efficiency. SFL is particularly well-suited for scenarios with (i) constrained communication bandwidth or asynchronous client participation, (ii) the absence of a trusted central server due to privacy or deployment restrictions, and (iii) client data exhibiting strong temporal dependencies. For large-scale models with tens to hundreds of billions of parameters, SFL alleviates bandwidth bottlenecks by reducing communication overhead [11], [12]. Its design philosophy parallels ring-all-reduce algorithms widely adopted in distributed training [13], [14], [15], highlighting the promise of sequential, low-communication strategies for scalable learning.

Under non-independent and identically distributed (non-IID) client data [6], [16], PFL aggregates client updates in parallel to mitigate heterogeneity, but diverse local distributions often induce conflicting update directions that steer the global model away from its optimum. In contrast, SFL adopts a chain-based, one-by-one update scheme that reduces such conflicts, yet its fully sequential nature frequently leads to unstable convergence and degraded performance. Existing proposed studies to address data heterogeneity have centered on PFL, including knowledge distillation [17], [18], [19], [20], [21], [22], loss regularization [23], [24], [25], [26], [27], similarity-based aggregation [28], [29], [30], [31], [32], and model decoupling [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], comparable solutions for SFL have yet to be explored.

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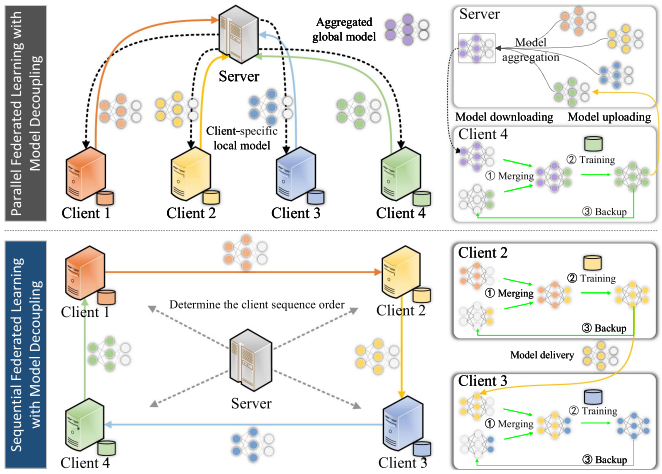


Fig. 1. Training manner for PFL and SFL with model decoupling.

We conduct motivation experiments across two real-world datasets and find that higher levels of data heterogeneity significantly amplify performance degradation between adjacent clients in sequential manner. Consequently, SFL demands to design a stabilization training technique to mitigate these performance drops during client-to-client model delivering. Model decoupling provides a promise to enhance training stability under non-IID conditions [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47]. Therefore, we consider designing customized model decoupling for SFL to mitigate the adverse effects of data heterogeneity by segmenting deep neural networks into a shared parameter subset for collaborative training and a personalized parameter subset for local heterogeneity adaptation. Fig. 1 illustrates the training paradigms of PFL and SFL with model decoupling. In SFL, clients share parameters sequentially, whereas PFL employs an all-gather approach. Due to the need to distribute the model among all clients, PFL incurs higher communication overhead. One of the most critical issues in model decoupling is determining which parameters should be personalized. To this end, we conducted motivation experiments by selecting a fixed number of model parameters from different locations as personalized, and found that the placement of personalized parameters significantly affects SFL training performance.

Several methodologies have been proposed for parameter personalization in PFL [41], [42], [47]. For example, works such as `FedSelect` [41] and `PSPFL` [47] propose identifying personalized parameters based on parameter gradients during local training, where larger update magnitudes indicate greater divergence between the local and global model. `FedDPA` [42] suggests utilizing parameters with higher Fisher information for personalization, which quantifies the change sensitivity of parameters to the loss function during local training. However, client interdependence complicates the separation of personalized and shared parameters in SFL, while gradient [41], [47] and Fisher [42]-based decoupling remains unexplored. Developing fine-grained element-wise scoring mechanisms is essential for effective parameter selection and stable, efficient sequential training.

In this paper, we propose `SeqFedEDT`, a novel element-wise decoupled training framework designed to accelerate SFL training by separating model parameters of each client into a shared subset for global knowledge collaboration and a personalized subset for migrating data heterogeneity. We utilize three types of parameter contribution scoring metrics based on gradient information, Fisher information, and parameter importance (PI), and employ these element-wise scoring methods to enable personalized model parameter selection. Moreover, we propose a novel quantile-based thresholding mechanism for separating shared and personalized parameter subsets, exploring optimal performance quantile selection through numerical studies. Extensive experimental validation across diverse datasets and heterogeneity scenarios demonstrates that `SeqFedEDT` consistently outperforms eight state-of-the-art (SOTA) methods, highlighting its superior performance and generalizability in handling SFL challenges. The main contributions of this work are as follows:

- Proposes a novel element-wise decoupled training framework, called `SeqFedEDT` that enhances SFL stability by adapting global knowledge to local data distributions. To the best of our knowledge, `SeqFedEDT` is the first to apply model decoupling personalized techniques in the SFL scenario.
- Introduces two parameter contribution scoring metrics for personalized parameter selection in SFL and proposes a novel fine-grained element-wise PI-based scoring method.
- Develops a quantile-based thresholding method using three element-wise contribution scores to separate shared and personalized model parameters. We design three practical SFL algorithms based on the three scoring metrics.
- Conducting extensive experiments, validating our solution performance superiority on diverse datasets and heterogeneity scenarios compared to three baselines and eight SOTA approaches.

The related works are reviewed in Section II. Section III presents the motivation experiments. Section IV provides system model and preliminaries of SFL. Section V gives the details of three scoring metrics and `SeqFedEDT` framework. Section VI details the algorithms' design and implementation. In Section VII, we evaluate our solution across diverse datasets and heterogeneity settings, comparing `SeqFedEDT` with three baselines and eight SOTA solutions. Section VIII concludes the paper and suggests future research directions.

II. Related Works

A. Personalized Federated Learning for non-IID Data

The traditional one-global-model FL paradigm, such as FedAvg, fails to perform effectively on non-IID data. To address this, Personalized FL (Per-FL) customizes models for each client based on local data distribution, mitigating statistical heterogeneity. We categorize mainstream Per-FL approaches into six technical routes. **Knowledge distillation-based** solutions [17], [18], [20], [21], [22] leverage knowledge transfer

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from historical global models or public datasets to alleviate client drift. However, their performance is critically dependent on the quality of the shared knowledge. **Multi-task learning-based** approaches [23], [24], [25], [26], [27] treat each client's local model as an independent task, capturing inter-task relationships through shared representations or joint regularization. Nonetheless, significant task differences or inaccurate affinity modeling can result in insufficient sharing or overfitting. **Clustering-based** methods [28], [29], [30], [31], [32] cluster clients based on similarity measures (e.g., model parameters or data distributions) and aggregate models within each cluster to enhance personalization. However, these methods incur substantial computational costs for similarity measurements and clustering. **Meta-learning-based** techniques [19], [48] aim to learn a global initialization or meta-parameters that can quickly adapt to client tasks with minimal local updates. Despite their effectiveness, they are sensitive to hyperparameters and may face convergence issues when client tasks exhibit significant variation. **Prototype-based** approaches [49], [50], [51] introduce category-level representation vectors (prototypes) that share semantic centers across clients, mitigating issues related to label shift or class imbalance. However, their effectiveness can be compromised when label space distributions are inconsistent or local classes are sparse.

Model Decoupling-based approaches [33], [34], [35], [36], [37], [38], [39], [40], [41], [42], [43], [44], [45], [46], [47] decompose the model into shared feature extractors and personalized output heads, enabling global knowledge sharing while allowing for local adaptation. `pFedFDA` [37] treats representation learning as a generative modeling task, where a global classifier is adaptively mapped to local feature distributions, enabling personalized model generation that balances global knowledge on non-IID data. `FedRoD` [27] introduces a dual-head model architecture, featuring a global head (G-Head) and a personalized head (P-Head). The G-Head captures common knowledge across clients, while the P-Head allows individual clients to adapt the model to their local data. However, most existing methods rely on static or pre-defined personalization strategies. In contrast, `SeqFedEDT` implements a Per-FL framework that integrates sequential model updates and fine-grained element-wise scoring to identify which parameters should be personalized. It reduces communication overhead in large-scale datasets and improves both adaptability and stability, particularly when clients exhibit high heterogeneity across tasks.

B. Sequential Federated Learning

SFL offers superior communication efficiency over PFL, making it a promising approach for large-scale distributed training [6]. However, current research reveals critical limitations in SFL, particularly in unstable convergence and performance degradation caused by client data heterogeneity [1], [2], [3], [4], [5], [6], [7], [8]. `TornadoAggregate` [1] employs a ring-based architecture and reformulates the loss minimization into a variance reduction to improve model accuracy and scalability. `FedSeq` [2] introduces the concept of "superclients", enabling sequential training of among subgroups of clients with heterogeneous data distributions to approximate PFL paradigm in a privacy-compliant way. `FedDC` [4] enhances SFL in data-sparse scenarios by periodically redistributing local models across clients in a privacy-preserving daisy chain manner. `Fed-CHS` [7] adopts a cluster-by-cluster sequential activation training paradigm, where clients within each cluster update the model sequentially, while parallel updates are carried out across different groups.

Most existing approaches [1], [2], [3], [4], [5], [6], [7], [8] combine sequential and parallel updates through client clustering—either sequentially within clusters and in parallel across clusters, or the reverse. Their objective remains the optimization of a single global model, leaving client-specific personalization largely unaddressed. In contrast, `SeqFedEDT` is designed to personalize models for each client and, to our knowledge, is the first framework in SFL to introduce parameter-level fine-grained model decoupling, enabling flexible adaptation to arbitrary client sequences. Furthermore, we introduce three element-wise scoring metrics to achieve stable training by retaining critical personalized parameters.

III. Motivation

Our motivation experiments comprise two main parts. First, we investigate SFL's accuracy variation with and without model decoupling. Second, we analyze accuracy results by decoupling model with different positions of personalized parameters. We conduct the experiments on datasets of *CIFAR10* and *CIFAR100* [52] and divided the datasets into 20 subsets, following the Dirichlet distribution *Dir*(α) [53] with α set to values within $\{0.1, 0.5, 1.0\}$, to simulate SFL training scenarios with non-IID data Distribution.

A. SFL Accuracy Variation With Non-IID Data

In Figure 2, we evaluate the training stability of standard SFL (black) and SFL with model decoupling (SFL-MD, red; classifier head fixed) under varying data heterogeneity. Each curve depicts client-level accuracy before and after local updates, with shaded regions indicating fluctuation ranges, while accompanying boxplots quantify iteration-wise accuracy variation. In Figure 2(a), client 7 on *CIFAR10* exhibits pronounced volatility under standard SFL, with the standard deviation σ (and mean absolute change $\overline{|\Delta|}$ per iteration) recorded as 24.97 (44.41), 12.43 (12.91), and 9.29 (14.27) for $\alpha \in \{0.1, 0.5, 1.0\}$, respectively. By contrast, SFL-MD markedly reduces fluctuations to 10.52 (2.57), 9.41 (2.23), and 4.89 (1.14). Similar patterns appear in Figures 2(b)–(d) on *CIFAR100* and client 0, thereby substantiating the instability of standard SFL and the stabilizing role of MD. Overall, across datasets, heterogeneity levels, and clients, MD consistently suppresses both variance and mean fluctuation magnitude, motivating its use as a decoupling mechanism to mitigate SFL training instability under non-IID data distributions.

Moreover, the results also reveal a nuanced effect of MD (with a fixed classifier head). Although MD stabilizes accuracy convergence, it can incur a drawback in low heterogeneity settings, where the final convergence accuracy tends to decline,

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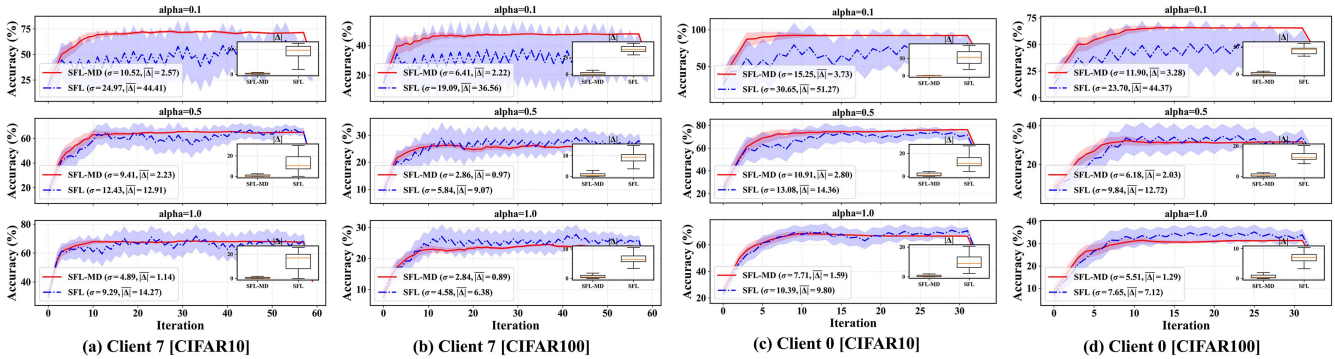


Fig. 2. Comparison of SFL and SFL with MD under Varying Data Heterogeneity ($\alpha = \{0.1, 0.5, 1.0\}$) and Datasets (*CIFAR10* and *CIFAR100*).

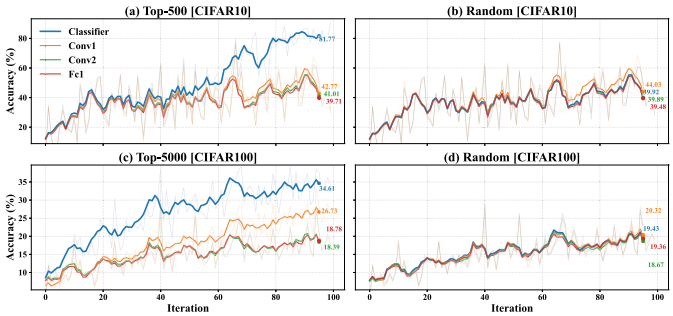


Fig. 3. Comparison of `Top-K` and `Random` Personalized Parameters Selection under Different Layers and Datasets.

as shown in Fig. 2. This degradation is especially pronounced on *CIFAR100*.

B. Evaluation of Personalized Parameter Selection

To understand the accuracy degradation caused by model decoupling, we investigated the impact of selecting personalized parameters at different positions for decoupling on the final accuracy performance. In our experiments, we tested the *CIFAR10* and *CIFAR100* datasets by selecting 500 personalized parameters at different positions with *CIFAR10* and 5000 personalized parameters with *CIFAR100*.

We adopted two selection strategies, including a `Random` strategy involving randomly selecting a certain number of personalized parameters, and a `Top-K` strategy prioritizing parameters with the largest divergence between global and local model parameters in SFL training. The faint thin curves represent the raw accuracy trajectories, while the bold solid curves denote the moving-average smoothed results for clearer visualization. Fig. 3(a)–(d) shows the results with the two strategies by selecting the parameters from different layers. Fig. 3(a) illustrates accuracy changes with iteration using top-500 parameters for model decoupling on the *CIFAR10* dataset. The results show that selecting personalized parameters from the *classifier* layer outperforms other layers (*conv1*, *conv2*, and *fc1*), while these layers exhibit nearly identical performance. A similar phenomenon is observed in Fig. 3(c), where experiments on the *CIFAR100* dataset selecting top 5000 parameters for personalization show that the *classifier* layer performs best, followed by *conv1*, while *conv2* and *fc1* demonstrate nearly comparable performance. These findings suggest that under the SFL training, selecting personalized parameters for model decoupling across different layers leads to performance disparities. Specifically, personalizing the *classifier* head consistently yields large gains relative to other layers.

Furthermore, Fig. 3(b) and (d) reveal that the `Random` strategy yields nearly identical accuracy when selecting personalized parameters from different layers, showing no significant performance variation between the layers. Although the `Top-K` strategy demonstrates a clear performance advantage when selecting personalized parameters from the *classifier* layer compared to other layers as shown in Fig. 3(a) and (c), the `Random` strategy fails to exhibit such a distinct pattern. This observation suggests that simply identifying a specific layer for personalization does not comprehensively resolve the parameter selection challenge. In addition, the results underscore that strategically choosing appropriate personalized parameters and their precise location remains critical for the SFL training.

IV. Preliminaries

In this section, we give the preliminaries of the SFL training [1], [2], [3], [4], [5], [6], [7], [8] and element-wise model decoupling in PFL [41], [42], [47]. For notation, scalars are represented by lowercase italic letters. Vectors and sets are denoted by lowercase bold italics and uppercase italics, respectively. Matrices are indicated with uppercase, italic, and bold fonts. The symbol Δ represents the differential operator, while $|\cdot|$ and $\|\cdot\|$ are used for the 1-norm and 2-norm, respectively. Table I lists the mathematical notations used in this paper.

A. Sequential Federated Learning

In the context of SFL, we denote $\mathcal{C}$ as the set of the clients, with $N = |\mathcal{C}|$ indicating the total number of clients. The i-th client is associated with a local dataset, denoted as $\mathcal{D}_i$. Since the data distributions are heterogeneous, every dataset $\mathcal{D}_i = \{\mathcal{X}_i, \mathcal{Y}_i\} = \{x_j^{(i)}, y_j^{(i)}\}_{j=1}^{d_i}$ is non-IID. Here, $\mathcal{X}_i$ and $\mathcal{Y}_i$ represent the set of input data and labels for $\mathcal{D}_i$ respectively, with d_i representing the data size of the i-th client where $d_i = |\mathcal{D}_i|$, x_j being the j-th input data, and y_j being the corresponding label. The total

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TABLE I
MATHEMATICAL NOTATIONS

Symbol	Description
$\mathcal{C}$	The set of the clients
N	The number of edge clients
$\mathcal{D}, \mathcal{D}_i$	The total dataset and the local dataset of i-th client
$\mathcal{X}_i, \mathcal{Y}_i$	The set of input data and label for i-th client respectively
d_i	The data size of the i-th client
d	The total number of samples across all clients
T	The set of communication rounds
K	The set of local update steps
$\mathcal{C}_t$	The set of participated clients in SFL at round t
n_t	The number of participated clients in SFL at round t
γ, α, η	The join ratio, data heterogeneity and learning ratio of clients
$\boldsymbol{o}_t$	The client indices order at round t
σ_j	The client index at position j in the order $\boldsymbol{o}_t$
$\boldsymbol{\theta}^k_{t,\sigma_j}$	The model of the client σ_j with k local training at round t
$\boldsymbol{\phi}^K_{t-1,\sigma_j}$	The personalized subset of the client σ_j at round $t-1$
$\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$	The shared subset of the client σ_{j-1} at round t
$\mathcal{M}(\boldsymbol{\theta})$	The set of all parameter indices in the set $\boldsymbol{\theta}$
$\mathbf{S}(\cdot)$	The parameter important scores for model decoupling
q, ρ	The quantile and threshold for element-wise decoupling

number of samples across all clients equals $d = \sum_{i\in\mathcal{C}} d_i$ and the total dataset is $\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\} = \{\mathcal{D}_i\}_{i\in\mathcal{C}}$.

We consider the SFL training process to encompass T communication rounds. In each round t, a subset of clients participates, defined by $\mathcal{C}_t \subset \mathcal{C}$, and the number of participated clients is denoted as $n_t = |\mathcal{C}_t| = \gamma \cdot |\mathcal{C}|$, where $t \in [0, T]$ and γ is the join ratio. Different from PFL, at the onset of each communication round, client indices order $\boldsymbol{o}_t = \{\sigma_1, \sigma_2, \ldots, \sigma_{n_t}\}$ are randomly sampled without replacement from the set $\{1, 2, \ldots, N\}$ to determine the sequential training order of the clients. At each round t, the model $\boldsymbol{\theta}_t$ is sequentially trained through clients according to the predetermined order $\boldsymbol{o}_t$. We denote $\boldsymbol{\theta}^k_{t,\sigma_j}$ as the model of the client σ_j with k local training at round t. Each client σ_j, $j \in \{1, 2, \ldots, n_t\}$ initializes its model parameters based on the trained model of the preceding client σ_{j-1}. For example, if σ_j is the first client in order $\boldsymbol{o}_t$, $\boldsymbol{\theta}^0_{t,\sigma_j} = \boldsymbol{\theta}^K_{t-1,\sigma_{n_t}}$ and for other cases $\boldsymbol{\theta}^0_{t,\sigma_j} = \boldsymbol{\theta}^K_{t,\sigma_{j-1}}$, where $\boldsymbol{\theta}^k_{t,i}$ denotes the local parameters of the i-th client after k local updates in the t-th round. The client σ_j performs K steps of local updates on its local dataset $\mathcal{D}_{\sigma_j}$ using stochastic gradient descent with learning rate η. After local training, the client σ_j passes the updated model $\boldsymbol{\theta}^K_{t,\sigma_j}$ to the next client σ_{j+1}. The round concludes when the final client σ_{n_t} completes its local training with model $\boldsymbol{\theta}^K_{t,\sigma_{n_t}}$.

The overall objective of SFL can be expressed as:

$$\arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i\in\mathcal{C}} f_i(\boldsymbol{\theta}), \tag{1}$$

where $f_i(\boldsymbol{\theta}) = \mathbb{E}_{(x,y)\sim\mathcal{D}_i}[\ell(\boldsymbol{\theta}; (x, y))]$, with $\ell(\cdot)$ denoting the local empirical loss function of i-th client.

B. SFL With Element-Wise Model Decoupling

Element-wise model decoupling addresses non-IID data by selecting a client-specific personalized subset $\boldsymbol{\phi}^K_{t-1,\sigma_j}$ from its local model parameters $\boldsymbol{\theta}^K_{t-1,\sigma_j}$ and a shared subset $\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$ from the updated model parameters $\boldsymbol{\theta}^K_{t,\sigma_{j-1}}$ passed by the preceding client. Client σ_j combines the personalized subset $\boldsymbol{\phi}^K_{t-1,\sigma_j}$ with the shared subset $\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$ to construct the merged model $\boldsymbol{\theta}^0_{t,\sigma_j} = \{\boldsymbol{\phi}^K_{t-1,\sigma_j}, \boldsymbol{\omega}^K_{t,\sigma_{j-1}}\}$. Let $\mathcal{M}$ denote the set of all parameter indices, such that:

$$\boldsymbol{\theta}^0_{t,\sigma_j} = \left\{\theta^{0,m}_{t,\sigma_j}\right\}_{m\in\mathcal{M}}, \tag{2}$$

where $\theta^{0,m}_{t,\sigma_j}$ represents the model parameter indexed by m for client σ_j at round t. For the subsets $\boldsymbol{\phi}^K_{t-1,\sigma_j}$ and $\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$, their corresponding parameter index sets are $\mathcal{M}(\boldsymbol{\phi}^K_{t-1,\sigma_j})$ and $\mathcal{M}(\boldsymbol{\omega}^K_{t,\sigma_{j-1}})$, respectively, as follows:

$$\boldsymbol{\phi}^K_{t-1,\sigma_j} = \left\{\theta^{K,m}_{t-1,\sigma_j} | m \in \mathcal{M}\left(\boldsymbol{\phi}^K_{t-1,\sigma_j}\right)\right\}, \tag{3}$$

$$\boldsymbol{\omega}^K_{t,\sigma_{j-1}} = \left\{\theta^{K,m}_{t,\sigma_{j-1}} | m \in \mathcal{M}\left(\boldsymbol{\omega}^K_{t,\sigma_{j-1}}\right)\right\}. \tag{4}$$

It should be noted that the element-wise model decoupling for each parameter m may vary across different clients and communication rounds t. Specifically, $\boldsymbol{\phi}^K_{t-1,\sigma_j}$ is updated solely using the client's local dataset $\mathcal{D}_{\sigma_j}$, whereas the $\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$ part is involved in both the local update and the SFL training of the preceding client's information.

V. FRAMEWORK WITH ELEMENT-WISE DECOUPLED TRAINING

In this section, we first present three element-wise importance metrics for model decoupling. Then, we provide the `SeqFedEDT` framework as applied in the SFL paradigm.

A. Element-Wise Importance Metrics

In traditional PFL model decoupling, the core issue is determining which parts are personalized parameters $\boldsymbol{\phi}^K_{t-1,\sigma_j}$ and which parts are shared parameters $\boldsymbol{\omega}^K_{t,\sigma_{j-1}}$. Researchers have currently proposed gradient-based scoring [41], [47] and Fisher information-based scoring [42] for element-wise model decoupling in PFL. In this subsection, we will introduce the gradient-based and Fisher information-based decoupling methods in SFL, and propose a novel decoupling scheme based on PI.

1) Gradient-Based Scoring: In SFL order $\boldsymbol{o}_t$, the preceding client σ_{j-1} delivers the trained model $\boldsymbol{\theta}^K_{t,\sigma_{j-1}}$ to the next client σ_j, which has the local model $\boldsymbol{\theta}^K_{t-1,\sigma_j}$ from the previous training round. The gradient-based scoring mechanism in SFL we propose is defined as the absolute difference between the trained model $\boldsymbol{\theta}^K_{t,\sigma_{j-1}}$ and the local model $\boldsymbol{\theta}^K_{t-1,\sigma_j}$ as follows:

$$\begin{aligned}\mathbf{S}_G\left(\theta^{K,m}_{t,\sigma_{j-1}}; \mathcal{D}_{\sigma_j}\right) &= \left|\nabla_{\theta^{K,m}_{t-1,\sigma_j}} \ell\left(\theta^{K,m}_{t-1,\sigma_j}; \mathcal{D}_{\sigma_j}\right)\right| \\ &= \left|\theta^{K,m}_{t,\sigma_{j-1}} - \theta^{K,m}_{t-1,\sigma_j}\right|,\end{aligned} \tag{5}$$

where $\mathbf{S}_G(\cdot)$ indicates the contribution of the parameter $\theta^{K,m}_{t-1,\sigma_j}$ in position m for personalization. This design differs from the scoring methods used in `FedSelect` [41] and `PSPFL` [47], which evaluate the parameter contribution by calculating the parameter difference between the pre-trained model $\boldsymbol{\theta}^0_{t,\sigma_j}$ and

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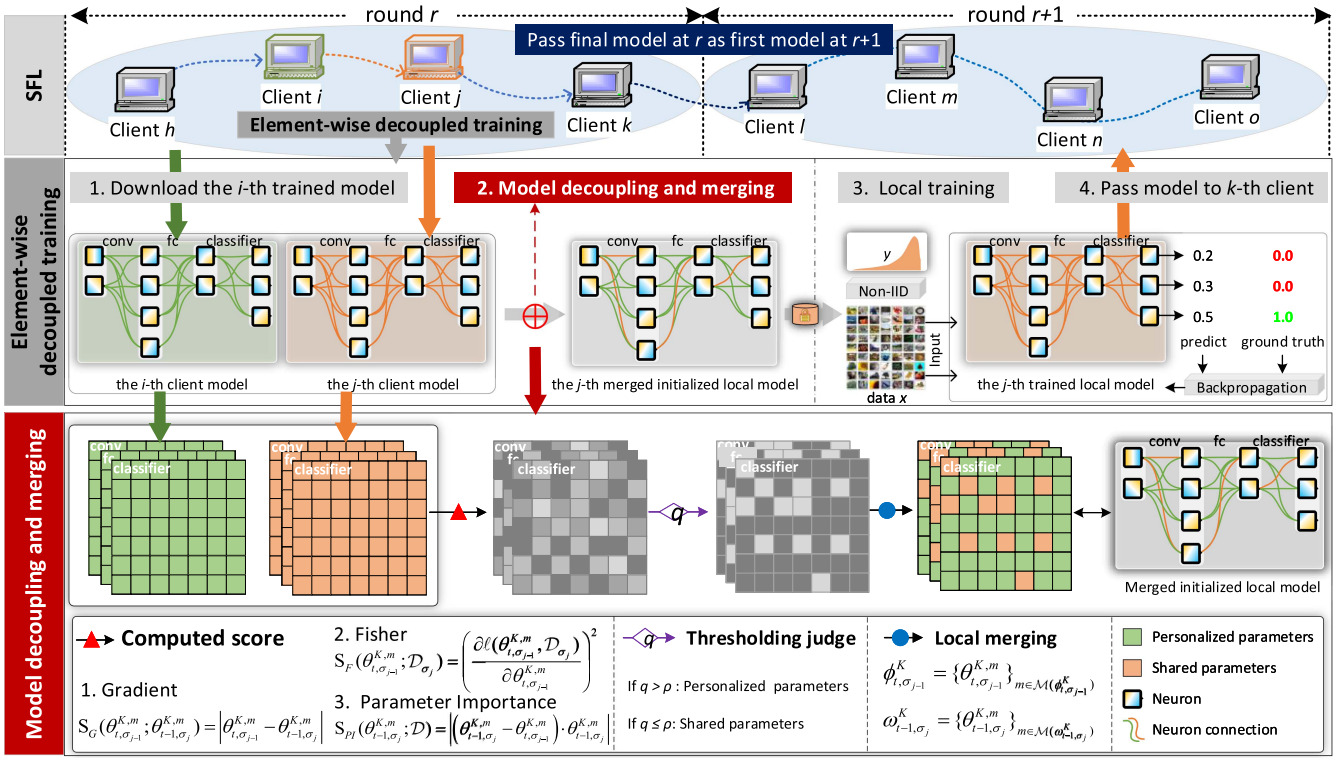


Fig. 4. The framework of SeqFedEDT with element-wise decoupled training and three types of important scores on non-IID data.

the post-trained model θ^{K}_{t,σ_j}. The larger $\mathbf{S}_G(\cdot)$, the stronger the parameter's effect on the current client's objective. Such parameters should be preserved for personalization to avoid damaging existing representations. Compared to (5), the previous designs can only be used if the client is selected again in subsequent rounds, introducing a delay issue when the participation rate $\gamma < 1$.

2) Fisher Information-Based Scoring: Fisher information serves as a measure of parameter sensitivity to changes in the dataset. FedDPA [42] utilizes Fisher information on the local dataset $\mathcal{D}_{\sigma_j}$ as the parameter contribution metric for personalization. In SFL, the scoring is calculated on model $\theta^{K,m}_{t,\sigma_{j-1}}$ before local training as:

$$\mathbf{S}_F\left(\theta^{K,m}_{t,\sigma_{j-1}};\mathcal{D}_{\sigma_j}\right) = \nabla_{\theta^{K,m}_{t,\sigma_{j-1}}}\ell\left(\theta^{K,m}_{t,\sigma_{j-1}};\mathcal{D}_{\sigma_j}\right)\cdot\nabla^{\top}_{\theta^{K,m}_{t,\sigma_{j-1}}}\ell\left(\theta^{K,m}_{t,\sigma_{j-1}};\mathcal{D}_{\sigma_j}\right), \quad (6)$$

A larger $\mathbf{S}_F(\cdot)$ suggests $\theta^{K,m}_{t,\sigma_{j-1}}$ is more sensitive to decision boundaries or output distribution, and retaining such parameters for personalization across clients helps stabilize training and alleviate prediction drift. The Fisher-based solution can be computed before local training, avoiding the delay issue present in the gradient-based method.

3) Parameter Importance-Based Scoring: We define a new scoring metric that measures the importance of a neural network parameter by evaluating the change in loss induced by removing the parameter. Inspired by prior studies [54], [55], we apply a Taylor expansion of the loss function at the m-th parameter and introduce a new element-wise importance score as follows:

$$\mathbf{S}_{PI}\left(\theta^{K,m}_{t-1,\sigma_j};\mathcal{D}\right) = \left|\theta^{K,m}_{t-1,\sigma_j}\cdot\nabla_{\theta^{K,m}_{t-1,\sigma_j}}\ell\left(\theta^{K,m}_{t-1,\sigma_j};\mathcal{D}\right)\right|, \quad (7)$$

$$\approx \left|\left(\theta^{K,m}_{t-1,\sigma_j}-\theta^{K,m}_{t,\sigma_{j-1}}\right)\cdot\theta^{K,m}_{t-1,\sigma_j}\right|. \quad (8)$$

where $\mathbf{S}_{PI}(\cdot)$ denotes the parameter importance of $\theta^{K,m}_{t-1,\sigma_j}$ for the personalization of client σ_j on $\mathcal{D}$. The approximation in (8) replaces the gradient term $\nabla_{\theta^{K,m}_{t-1,\sigma_j}}\ell(\theta^{K,m}_{t-1,\sigma_j};\mathcal{D})$ with the parameter update $(\theta^{K,m}_{t-1,\sigma_j}-\theta^{K,m}_{t,\sigma_{j-1}})$ in (7). Preserving high-PI parameters maintains backbone integrity and prevents cumulative errors and long-term degradation.

B. SeqFedEDT Framework

The proposed SeqFedEDT framework is illustrated in Fig. 4. The SFL process can be seen as a multi-round sequential model training. In each round t, the server determines the sequential order $\boldsymbol{o}_t$ and the model ϕ^0_{t,σ_1} is consecutively trained on each client, which involves four key steps:

- *Model Retrieval from Previous Client:* If the client is the first client ($\sigma_j = 1$), it directly receives the model ϕ^0_{t,σ_1}. For other conditions, the client $\sigma_j > 1$ obtains a new updated model $\phi^0_{t,\sigma_{j-1}}$ from the previous client σ_{j-1} in the order $\boldsymbol{o}_t$.
- *Model Decoupling and Merging:* The process consists of three main steps, as described in detail below:

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- *Element-wise Importance Scoring:* The client σ_t calculates the parameter score $\mathbf{S}(\cdot)$ by selecting one of the three scoring methods outlined in (5), (6), and (7), respectively.
- *Threshold-based Decoupling:* Importance-based partitioning employs a quantile-based thresholding function $f_q : \mathbf{S}(\cdot) \to \mathbb{R}$ to determine a threshold ρ as follows:

$$\begin{aligned}\rho &= f_q(\mathbf{S}(\cdot)) \\ &\triangleq \min_{\mathbf{s}\in\mathbb{R}} \{\mathbf{s} : \mathbb{E}\left[\mathbf{1}\left(\mathbf{S}(\cdot) \le \mathbf{s}\right)\right] \ge q\}, \end{aligned} \tag{9}$$

where $q \in [0, 1]$ denotes the desired quantile level, and $\mathbb{E}[\mathbf{1}(\mathbf{S}(\cdot) \le \mathbf{s})]$ corresponds to the cumulative probability of $\mathbf{S}(\cdot)$ not exceeding $\mathbf{s}$. Then, the client σ_t performs ρ-threshold-based decoupling as follows:

$$\mathcal{M}\left(\boldsymbol{\phi}^0_{t-1,\sigma_j}\right) = \left\{m \mid \mathbf{S}(\theta^{K,m}_{t-1,\sigma_j}) > \rho\right\}, \tag{10}$$

$$\mathcal{M}\left(\boldsymbol{\omega}^0_{t,\sigma_{j-1}}\right) = \left\{m \mid \mathbf{S}(\theta^{K,m}_{t-1,\sigma_j}) \le \rho\right\}. \tag{11}$$

- *Model Merging:* The client σ_j obtains the merged model $\boldsymbol{\theta}^0_{t,\sigma_j}$ by combining the client-specific personalized parameter subset $\boldsymbol{\phi}^0_{t,\sigma_j}$ with the shared parameter subset $\boldsymbol{\omega}^0_{t,\sigma_j}$ as:

$$\boldsymbol{\theta}^0_{t,\sigma_j} = \boldsymbol{\phi}^0_{t-1,\sigma_j} \cup \boldsymbol{\omega}^0_{t,\sigma_{j-1}}. \tag{12}$$

- *Local Model Training:* client σ_j performs local training on the merged model $\boldsymbol{\theta}^0_{t,\sigma_j}$ with the local dataset $\mathcal{D}_{\sigma_j}$ to obtain the trained model $\boldsymbol{\theta}^K_{t,\sigma_j}$ as follows:

$$\boldsymbol{\theta}^{k+1}_{t,\sigma_j} = \boldsymbol{\theta}^k_{t,\sigma_j} - \eta \nabla_{\boldsymbol{\theta}} \ell\left(\boldsymbol{\theta}^k_{t,\sigma_j}; \mathcal{D}_{\sigma_j}\right), \tag{13}$$

where η is the learning rate, and the update can be performed multiple times.
- *Next-hop Model Delivery:* If the current client is the last client σ_{n_t}, the order $\boldsymbol{o}_{t+1}$ needs to be regenerated. Otherwise, the client σ_j delivers its locally trained model $\boldsymbol{\theta}^K_{t,\sigma_j}$ to the next client σ_{j+1}.

VI. Algorithm Design

In this section, we first present SeqFedEDT algorithms. Then, we provide the implementation and complexity analysis of the algorithms.

A. SeqFedEDT Algorithms

The SeqFedEDT framework is implemented through two main algorithms, as illustrated in Algorithms 1 and 2. Algorithm 2 acts as a subroutine of Algorithm 1, providing the execution process for each clients in the SFL workflow.

Algorithm 1 outlines how clients are sequentially organized for model training across multiple rounds. During each training round t, the server samples a permutation $\boldsymbol{o}_t$ of the client indices to establish the order in which clients will participate sequentially in the training process. Each client, represented by σ_i, retrieves the model based on its position in the sequence. If a client is the first in line, it utilizes the pre-round last model as initialization, whereas subsequent clients invoke the Element-wise Decoupled Training $\texttt{EDT}(\cdot)$ function using the updated model received from the preceding client. This iterative process continues until all clients have completed their training, resulting in a set of client-specific trained models $\{\boldsymbol{\theta}_i\}_{i\in\mathcal{C}}$.

Algorithm 1: Sequential Federated Learning.

Input: Number of rounds T, the client set $\mathcal{C}$, and the number of local steps K.
Output: The client-specific trained models $\{\boldsymbol{\theta}_i\}_{i\in\mathcal{C}}$.

```
Initialize θ_0^0 ;
for each training round t = 0, 1, ..., T − 1 do
    Sample a permutation o_t = {σ_1, σ_2, ..., σ_{n_t}}
      from the client index set {1, 2, ..., N} ;
    for in sequence o_t = {σ_1, σ_2, ..., σ_{n_t}} do
        θ^K_{t,σ_i} = { EDT(θ^K_{t-1}),          if i = 1
                      { EDT(θ^K_{t,σ_{i-1}}),    if i > 1
    end
end
return {θ_i}_{i∈C} ;
```

Algorithm 2, named EDT, provides a focused implementation for each client's training procedure within the SFL framework. This algorithm begins by initializing the local model $\boldsymbol{\theta}^K_{t-1,\sigma_j}$, the training epochs K, the local dataset $\mathcal{D}_{\sigma_j}$, and the learning rate η. It start computes an element-wise importance score $\{\mathbf{S}(\theta^{K,m}_{t-1,\sigma_j})\}$ for the model parameters, employing one of three established scoring methods based on (5), (6), or (7) to identify which parameters should be personalization. Following this, the algorithm decouples the parameters into personalized and shared components, denoted as $\boldsymbol{\phi}^0_{t-1,\sigma_j}$ and $\boldsymbol{\omega}^0_{t,\sigma_{j-1}}$ based on the importance scores. The merging process combines the personalized parameters of the current client with the parameters from the preceding client, forming a merged model for training as $\boldsymbol{\theta}^0_{t,\sigma_j} = \boldsymbol{\phi}^0_{t-1,\sigma_j} \cup \boldsymbol{\omega}^0_{t,\sigma_{j-1}}$. The EDT algorithm subsequently enters a training loop, where the client iteratively updates its model using the local dataset $\mathcal{D}_{\sigma_j}$, employing the specified learning rate η. Upon completion of the allocated training epochs, the updated model $\boldsymbol{\theta}^K_{t,\sigma_j}$ is returned to Algorithm 1 for further progression in the SFL training cycle.

B. Implementation and Complexity Analysis

EDT Algorithm has a time complexity of $\mathcal{O}(K \cdot |\mathcal{M}|)$, where K is the number of training epochs and $|\mathcal{M}|$ is the number of model parameters. This complexity arises from computing element-wise importance scores, decoupling parameters, merging models, and performing local updates over K epochs. The space complexity is $O(|\mathcal{M}|)$, primarily for storing the model parameters and importance scores. In contrast, Algorithm 1, which orchestrates the training of multiple clients, exhibits a time complexity of $O(T \cdot N \cdot K \cdot |\mathcal{M}|)$, where T is the number of training rounds and N is the total number of clients. This is due to invoking Algorithm 1 for each client across all training

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Algorithm 2: Element-wise Decoupled Training.

Init: The local model $\boldsymbol{\theta}^K_{t-1,\sigma_j}$, the training epoch K, the local dataset $\mathcal{D}_{\sigma_j}$, and learning rate η.
Input: The preceding model $\boldsymbol{\theta}^K_{t,\sigma_{j-1}}$.
Output: The updated model $\boldsymbol{\theta}^K_{t,\sigma_j}$.

Compute element-wise importance score $\{\mathbf{S}(\theta^{K,m}_{t-1,\sigma_j})\}$ based on Eqs. (5), (6), or (7) ;
Decouple $\boldsymbol{\phi}^0_{t-1,\sigma_j}$ and $\boldsymbol{\omega}^0_{t,\sigma_{j-1}}$ based on Eqs. (10), (11) ;
Merge model $\boldsymbol{\theta}^0_{t,\sigma_j} = \boldsymbol{\phi}^0_{t-1,\sigma_j} \cup \boldsymbol{\omega}^0_{t,\sigma_{j-1}}$;
for $k = 0$ **to** $K-1$ **do**
  Train $\boldsymbol{\theta}^{k+1}_{t,\sigma_j} \leftarrow \boldsymbol{\theta}^k_{t,\sigma_j} - \eta \nabla_\theta \ell(\boldsymbol{\theta}^k_{t,\sigma_j}; \mathcal{D}_{\sigma_j})$;
end
return $\boldsymbol{\theta}^K_{t,\sigma_j}$;

rounds. Here, $|\mathcal{M}|$ can be computed in parallel, mitigating potential computational bottlenecks. Meanwhile, since each client's training instance is independent, the workload can be distributed efficiently across available resources.

VII. Performance Evaluation

In this section, we outline the experimental settings, datasets, baseline comparisons, and implementation details. We assess the performance of our solution by benchmarking it against eight SOTA methods, focusing on accuracy, convergence, score comparison, scalability, ablation study, communication overhead and of personalized parameter placement analysis.

A. Experiment Setup and Datasets

We build the experimental prototype based on an open-source benchmark.[1] All experiments are conducted on a high-performance server with GPU (NVIDIA GeForce RTX 4090 24 GB) and CPU (Intel i9-13900 K, 24-Core, 3.00 GHz), using Python 3.10.13. The code is available at repository[2].

1) Datasets: Our experiments encompass eight open benchmark datasets including *CIFAR10*, *CIFAR100*, *EMNIST*, *FMNIST*, *MEDMNISTA*, *MEDMNISTC*, *MNIST*, and *SVHN*. For further details, please refer to Appendix A.2. Each dataset was partitioned into 100 subsets ($N = 100$), following the Dirichlet distribution *Dir*(α) as established in [53], with α set to values within $\{0.1, 0.5, 1.0\}$, which was employed to simulate scenarios on non-IID data. The degree of heterogeneity is determined by α, where a smaller α value indicates more significant data heterogeneity. For instance, when $\alpha = 0.1$, the degree of non-IID data exhibits a high level of data heterogeneity. This observation confirms that the dataset assigned to any individual client does not include all available classes (*i.e.*, $|\mathcal{Y}_i| < |\mathcal{Y}|$). Here, $\mathcal{Y}_i$ denotes the label space of the data allocated to the i-th client.

2) Baselines: We compare `SeqFedEDT` against eight SOTA FL solutions. To establish baselines for evaluating generalization and personalization performance, we implement `SFL` [6], `FedAvg` [9], and `Local-only` in our experiments. The `Local-only` approach indicates that clients update parameters solely using their local data without any global parameter aggregation. Furthermore, we present accuracy comparisons for the eight algorithms detailed hyper-parameter settings can be found in Appendix A.4.

- `APFL` [43] enables clients to train local models while adaptively integrating local and global parameters to contribute to the global model.
- `FedPer` [33] utilizes last-layer-as-head decoupling by splitting the network into the base and personalized layer to achieve personalization.
- `FedProto` [45] is a prototype aggregation framework for FL that addresses heterogeneous clients by exchanging class prototypes instead of gradients, mitigating optimization convergence issues arising from data distribution differences and model architecture heterogeneity.
- `FedRep` [34] treats the classifier as the personalized layer, and trains the classifier locally and the feature extractor sequentially.
- `Floco` [46] leverages linear mode connectivity to identify a linearly connected low-loss region within the parameter space of neural networks. This method enables clients to personalize their local models within specific subregions while collaborating to train a global model.
- `LG-FedAvg` [44] adopts an alternative approach by training the feature extractor locally while training the classifier globally to alleviate the effects of data heterogeneity.
- `pFedFDA` [37] addresses the limitations of transitional layer-wise model decoupling, particularly the bias-variance trade-off in classifier training that depends solely on local datasets. Additionally, it conceptualizes classifier representation learning as a generative modeling task, training representations based on the global feature distribution.

3) Implementation: We use a simple *AvgCNN* model with 2 convolution layers and 2 fully connected layers for *CIFAR10*, *CIFAR100*, *EMNIST*, *FMNIST*, *MNIST*, and *SVHN* and *LeNet5* [56] for *MEDMNISTA*, and *MEDMNISTC* across all methods to conduct performance evaluations. Following the setup in [39], we evenly partition the dataset of each edge client into training and test sets with no overlap, *i.e.*, $|\mathcal{D}_i^{train}| = |\mathcal{D}_i^{test}|$. On the edge client side, we adopt stochastic gradient descent as the local optimizer. The local learning rate is set to $\eta = 0.01$. All experiments shared the same join ratio $\gamma = 0.1$, communication round $T = 400$, local epoch $K = 5$, and batch size equals to 32. In Appendix A, we provide the list of the model architectures and the full hyperparameter configurations employed in the methods involved. We provide `SeqFedEDT` with three types of scoring methods, including `Grad.`, `Fisher`, and `PI`, respectively.

B. Comparison of Classification Accuracy

We compared the average accuracy of our solution with other baselines at $\alpha = 0.1$, as shown in Table II. Additional results

[1]https://github.com/KarhouTam/FL-bench
[2]https://github.com/tian0920/SeqFedEDT

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TABLE II
AVERAGE ACCURACY WITH STANDARD DEVIATION OVER DATASETS FOR $\alpha = 0.1$. BOLD AND UNDERLINED INDICATE THE BEST AND SECOND-BEST, RESPECTIVELY

Method	CIFAR10	CIFAR100	EMNIST	FMNIST	MEDMNISTA	MEDMNISTC	MNIST	SVHN
FedAvg	58.01 ± 0.05	25.99 ± 0.14	82.16 ± 0.10	87.03 ± 0.03	18.10 ± 0.00	9.03 ± 0.00	98.51 ± 0.00	87.78 ± 0.08
SFL	65.57 ± 0.00	29.74 ± 0.18	78.88 ± 0.14	87.88 ± 0.00	18.10 ± 0.00	22.28 ± 0.00	99.00 ± 0.00	90.99 ± 0.01
Local-only	80.80 ± 0.00	37.59 ± 0.03	93.18 ± 0.01	94.88 ± 0.02	67.29 ± 0.07	66.67 ± 0.20	97.46 ± 0.00	90.00 ± 0.01
APFL	58.66 ± 0.09	26.30 ± 0.18	81.95 ± 0.09	89.08 ± 0.02	63.32 ± 0.11	64.51 ± 0.06	98.87 ± 0.01	89.09 ± 0.08
FedPer	84.73 ± 0.02	41.82 ± 0.14	94.19 ± 0.01	96.12 ± 0.01	67.34 ± 0.00	66.53 ± 0.00	99.00 ± 0.00	94.60 ± 0.03
FedProto	82.54 ± 0.01	42.06 ± 0.04	93.99 ± 0.00	95.21 ± 0.00	67.10 ± 0.01	66.53 ± 0.00	97.79 ± 0.00	91.34 ± 0.02
FedRep	84.41 ± 0.01	39.15 ± 0.06	94.23 ± 0.01	95.68 ± 0.01	67.00 ± 0.11	66.01 ± 0.02	98.53 ± 0.00	94.03 ± 0.00
FedSeq	63.40 ± 0.01	27.93 ± 0.01	74.33 ± 0.00	86.48 ± 0.01	11.16 ± 0.00	22.28 ± 0.00	98.80 ± 0.01	90.36 ± 0.02
FLOCO	80.46 ± 0.58	27.71 ± 0.48	88.74 ± 0.07	94.72 ± 0.02	58.71 ± 0.97	57.60 ± 0.55	98.70 ± 0.02	93.07 ± 0.02
LG-FedAvg	81.59 ± 0.04	38.65 ± 0.02	93.64 ± 0.01	95.20 ± 0.00	67.34 ± 0.00	66.56 ± 0.00	97.74 ± 0.00	91.44 ± 0.02
pFedFDA	86.37 ± 0.04	41.98 ± 0.06	93.32 ± 0.00	96.00 ± 0.00	67.45 ± 0.02	66.78 ± 0.01	98.73 ± 0.01	93.87 ± 0.02
SeqFedEDT (Grad.)	87.51 ± 0.14	51.23 ± 0.64	94.13 ± 0.06	96.99 ± 0.01	67.25 ± 0.32	66.78 ± 0.16	**99.46 ± 0.02**	95.60 ± 0.02
SeqFedEDT (Fisher)	84.82 ± 0.22	45.59 ± 0.13	93.51 ± 0.05	96.58 ± 0.09	67.21 ± 0.16	66.75 ± 0.06	99.35 ± 0.00	95.03 ± 0.05
SeqFedEDT (PI)	**88.21 ± 0.08**	**52.61 ± 0.08**	**94.61 ± 0.02**	**97.09 ± 0.04**	**67.47 ± 0.03**	**66.88 ± 0.07**	99.44 ± 0.00	**95.87 ± 0.01**

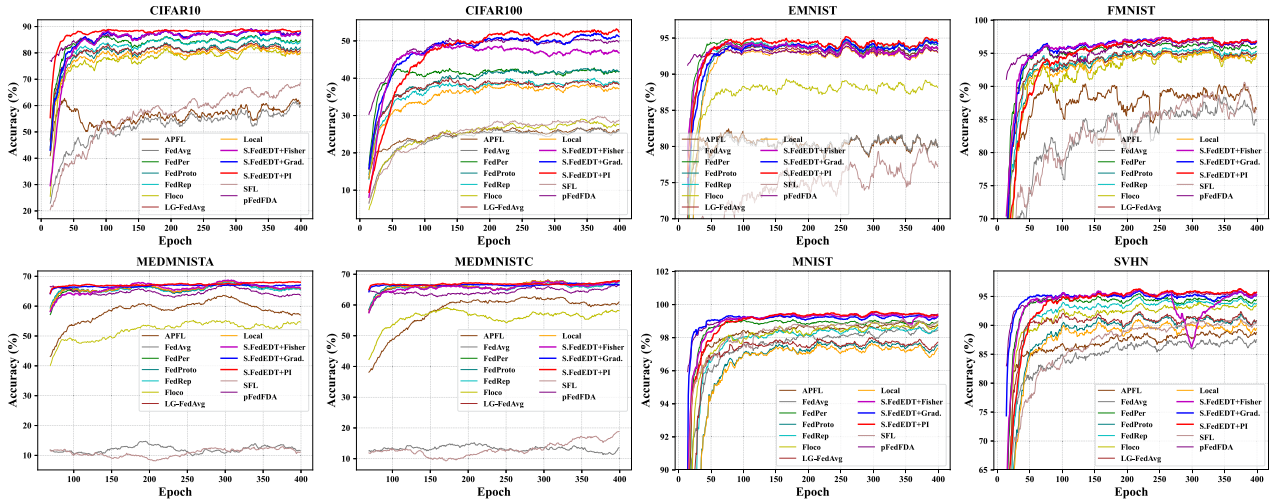


Fig. 5. Convergence comparison of SeqFedEDT against three benchmarks and eight SOTA solutions with $\alpha = 0.1$.

(under $\alpha = \{0.5, 1.0\}$) are provided in Appendix B.1. The results indicate that SeqFedEDT excels across most datasets, consistently achieving Top-2 performance and often reaching the highest accuracy. Specifically, the average accuracy in the datasets *CIFAR10*, *CIFAR100*, *EMNIST*, *FMNIST*, *MEDMNISTA*, *MEDMNISTC*, *MNIST* and *SVHN* is 88.21%, 52.61%, 94.61%, 97.09%, 67.47%, 66.88%, 99.46% and 95.87%, respectively (the best overall). Notably, SeqFedEDT demonstrates a significant lead in the complex dataset *CIFAR100*, with an accuracy improvement of 10.55% over other PFL methods. Additionally, the SeqFedEDT with PI method consistently achieves optimal or near-optimal results compared to the SeqFedEDT with Grad. and Fisher based methods, further demonstrating the performance advantages of PI-based method. Considering Tables 13 and 14, SeqFedEDT illustrates stable performance across different values of α, highlighting its flexibility and robustness in handling non-IID data.

C. Convergence Analysis

Fig. 5 provides the convergence performance of all solution across eight datasets with $\alpha = 0.1$. We use a bold line to highlight the series of SeqFedEDT algorithms with three types of scores. On *CIFAR10*, SeqFedEDT with PI achieves the highest accuracy and exhibits the fastest convergence, while SeqFedEDT with Grad. and Fisher also attain comparable performance. pFedFda, FedRep and FedPer also show good convergence reaching above 84%. Other methods, such as FedProto, FLOCO and LG-FedAvg, exhibit slower convergence, within the accuracy range of approximately 80% to 83%. In other datasets, at least one of the SeqFedEDT algorithms demonstrates both the fastest convergence and top-2 performance among all alternative solutions. In addition, SeqFedEDT requires only approximately 100 rounds to converge across all dataset scenarios, while SFL typically requires around 250 rounds. Notably, in the case of *FMNIST*, SFLalgorithm takes

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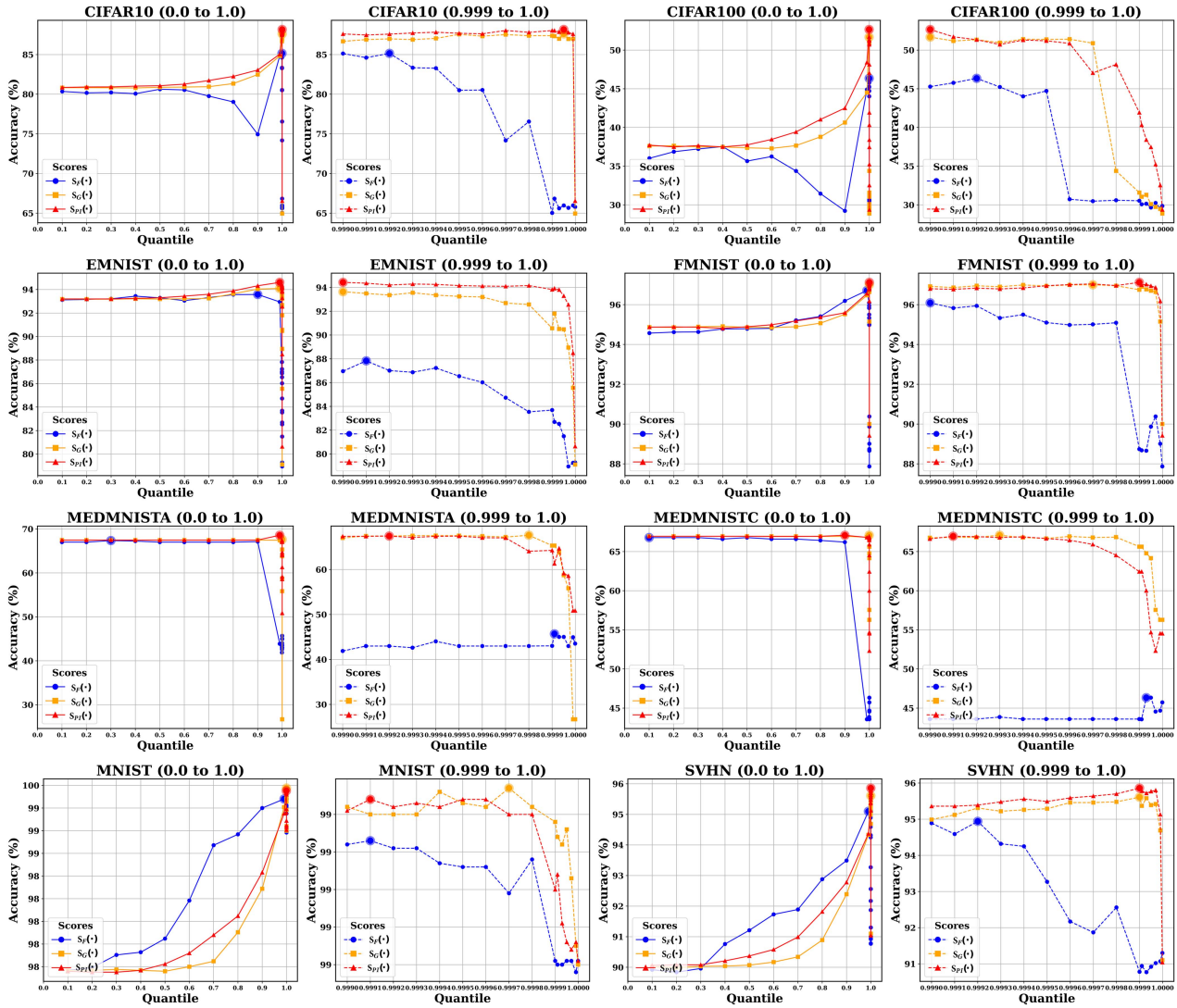


Fig. 6. Comparison of three score types, including Gradient $\mathbf{S}_G(\cdot)$, Fisher information $\mathbf{S}_F(\cdot)$, and PI score $\mathbf{S}_{PI}(\cdot)$, across eight datasets.

roughly 350 rounds to reach stability. This significant difference in convergence speed further highlights the effectiveness of the `SeqFedEDT` methodology compared to original SFL.

Figs. 8 and 9 (see Appendix B.2) show similar trends. In summary, `SeqFedEDT` algorithms achieve competitive convergence performance across all datasets, with a notable acceleration observed in SFL training. This improvement is attributed to element-wise decoupling guided by three scoring metrics, which separates personalized from shared parameters and enhances convergence.

D. Score Comparison

Through numerical experiments, we evaluated the performance of three distinct scoring methods (`Grad.` $\mathbf{S}_G(\cdot)$, `Fisher` $\mathbf{S}_F(\cdot)$, and `PI` $\mathbf{S}_{PI}(\cdot)$). We tested these scoring methods across quantile q configurations ranging from 0.0 to 1.0. A higher quantile corresponds to a reduced proportion $(1-q)$ of personalized parameters. The experimental setup employs the *LeNet5* architecture for the *MEDMNISTA* and *MEDMNISTC* datasets, while utilizing a *AvgCNN* model for the other six datasets. We consistently set the total communication rounds to $K = 400$ and maintain a default γ value of 0.1.

Across datasets, increasing the quantile q reduces personalized parameters $\boldsymbol{\phi}^0_{t-1,\sigma_j}$ and increases shared parameters $\boldsymbol{\omega}^0_{t,\sigma_{j-1}}$, causing the three scores to initially rise and subsequently decline. Each score exhibits a different peak performance at different quantile q values across datasets. As shown in Fig. 6, $\mathbf{S}_{PI}(\cdot)$ provides the best performance on almost all datasets except *MNIST* compared to $\mathbf{S}_G(\cdot)$ and $\mathbf{S}_F(\cdot)$. For examples, $\mathbf{S}_G(\cdot)$ reaches peak at quantile about $q = 0.999$, $\mathbf{S}_F(\cdot)$ and $\mathbf{S}_{PI}(\cdot)$ peaks at quantile about $q = 0.99995$ with accuracy above 85% on *CIFAR10*. These results indicate that to achieve optimal performance, $\mathbf{S}_G(\cdot)$ requires approximately 879 personalized parameters, $\mathbf{S}_F(\cdot)$ and $\mathbf{S}_{PI}(\cdot)$ need only 44. On CIFAR100, $\mathbf{S}_G(\cdot)$ reaches peak at about $q = 0.9991$ (833), $\mathbf{S}_F(\cdot)$ and $\mathbf{S}_{PI}(\cdot)$ peak at quantile about $q = 0.999$ (925). Notably, across the other six datasets, a consistent trend emerges where $\mathbf{S}_{PI}(\cdot)$ outperforms both $\mathbf{S}_G(\cdot)$ and $\mathbf{S}_F(\cdot)$, with the sole exception of *MNIST*, where `PI` performance falls slightly below `Fisher`

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TABLE III
ABLATION STUDY OF SeqFedEDT WITH BENCHMARKS ON EIGHT DATASETS

Dataset	FedAvg	Local	SFL	S.FedEDT†	S.FedEDT
CIFAR10	58.04	80.80	65.57	64.82	**88.16**
CIFAR100	25.89	37.61	29.87	38.99	**52.66**
EMNIST	82.23	93.19	78.78	94.26	**94.61**
FMNIST	87.05	94.87	87.88	90.18	**97.12**
M.MNISTA	18.10	67.34	18.10	64.69	**68.57**
M.MNISTC	9.03	66.51	22.28	66.73	**67.08**
MNIST	98.51	97.46	99.00	99.03	**99.44**
SVHN	87.83	90.01	91.00	91.29	**95.86**

TABLE IV
PROPORTION OF SELECTED PERSONALIZED PARAMETERS ACROSS DIFFERENT LAYERS WITH PI SCORE AND ABSOLUTE COUNTS IN $(\cdot)$

Dataset	Conv	FC	Classifier
CIFAR100	0.136 (6)	0.002 (1)	0.723 (32)
CIFAR10	0.277 (256)	0.001 (1)	0.443 (410)
EMNIST	0.135 (576)	0.059 (252)	0.613 (2,612)
FMNIST	0.342 (20)	0.001 (1)	0.315 (19)
M.MNISTA	0.042 (187)	0.404 (1,799)	0.120 (534)
M.MNISTC	0.076 (339)	0.207 (922)	0.489 (2,177)
MNIST	0.277 (146)	0.002 (2)	0.442 (232)
SVHN	0.242 (22)	0.001 (1)	0.513 (46)

scoring method. These results reveal that three scoring methods facilitate parameter decoupling in SFL training, ensuring stable convergence and high accuracy across datasets. Notably, optimal performance is often achieved with only a minimal fraction of personalized parameters for all datasets.

E. Layer-Wise Parameter Distribution With PI Score

To further examine PI's layer-level behavior, Table IV reports the proportion of personalized parameters selected across convolutional (*conv*), fully connected (*fc*), and *classifier* layer. For datasets employing the *AvgCNN* backbone (all except *MEDMNISTA/C*), selected personalized parameters are predominantly concentrated in the *classifier* layer, accounting for 44.3% 72.3%, whereas contributions from *conv* and *fc* layers remain minor. In contrast, *MEDMNISTA/C*, which adopts the *LeNet5* backbone, exhibits a distinct pattern: its shallower architecture and higher *fc* parameter proportion lead PI to preferentially select more personalized parameters in the *fc* layers. These findings suggest that PI's selection preference is not intrinsic but largely dictated by the underlying backbone's parameter distribution: when the *classifier* layer dominates, it is prioritized, whereas in architectures with *fc* layers, PI favors them instead.

F. Ablation Study

We conducted ablation experiments on *AvgCNN* and *ResNet18* (see Appendix B.3) to evaluate the performance of SeqFedEDT using the PI score. SeqFedEDT† is a variant of our solution which only selects the personalized parameters from the classifier layer.

Table III presents the accuracy results across eight datasets, for variants of SeqFedEDT, comparing them with the basic FedAvg, Local-only, and the original SFL method. As shown in Table III, for *CIFAR10*, our method achieved an accuracy of 88.16%, significantly surpassing the layer-wise SeqFedEDT† method (64.82%). Similarly, the results on other datasets such as *CIFAR100*, *MNIST*, and *SVHN* show SeqFedEDT outperformed all other approaches, and SeqFedEDT† often provides better performance than FedAvg and SFL. Fig. 10 (in Appendix B.3) shows the convergence analysis under the same conditions. For all datasets, SeqFedEDT provides better performance than SeqFedEDT† and SeqFedEDT variants achieve faster convergence than FedAvg and SFL benchmarks. Although the Local-only method performs well, it is consistently lower than SeqFedEDT. The results indicate that sequential training consistently outperforms parallel aggregation, and applying parameter decoupling to the full model is more effective than limiting it to the classifier alone. Most importantly, the ablation study highlights the significant potential of element-wise decoupling within SFL training.

TABLE V
ACCURACY OF SeqFedEDT ON EIGHT DATASETS WITH DIFFERENT JOIN RATIOS

Dataset	0.1	0.2	0.3	0.4	0.5
CIFAR10	88.16	87.96	88.20	**88.34**	88.14
CIFAR100	**52.66**	52.42	52.35	52.49	52.01
EMNIST	94.61	94.64	**94.70**	94.67	**94.70**
FMNIST	**97.12**	97.05	97.07	97.11	97.10
M.MNISTA	67.30	67.33	67.36	67.48	**68.57**
M.MNISTC	66.56	66.96	66.67	66.80	**67.08**
MNIST	**99.44**	**99.44**	99.40	**99.44**	**99.44**
SVHN	**95.86**	95.79	95.75	95.85	95.84

G. Scalability

The performance of our SeqFedEDT method was evaluated under different client join ratios $\gamma = \{0.1, 0.2, 0.3, 0.4, 0.5\}$ across eight diverse datasets. As shown in Table V, the results reveal distinct performance patterns across the datasets. For *CIFAR10*, performance remains relatively stable, with a slight peak at $\gamma = 0.4$ (88.34%), indicating our solution has minimal sensitivity to client participation. This phenomenon is also observed in other datasets. In *CIFAR100*, our solution's accuracy remains around 52-53% with different joint ratio, peaking at 52.66% when $\gamma = 0.1$. The counterintuitive result stems from the fact that higher participation in SFL prolongs the update sequence, intensifying error propagation and positional bias, ultimately impairing performance on complex datasets. In the case of *MNIST*, accuracy consistently reaches 99.44% across different join ratios, indicating that the model is highly resilient to variations in client participation. Overall, SeqFedEDT exhibits robust performance across the datasets, with certain datasets benefiting from increased client participation, particularly in the *MEDMNIST* variants.

H. Communication and Computation Overhead

1) Communication Overhead: Table VI shows the computational and communication costs of SeqFedEDT compared to several existing approaches on *CIFAR10*. For computational

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TABLE VI
THE COMPUTATIONAL COST IS EVALUATED ON *CIFAR10*, AND THE COMMUNICATION COST IS DEFINED AS THE NUMBER OF PARAMETERS TRANSMITTED PER ITERATION. Σ DENOTES THE TOTAL NUMBER OF PARAMETERS OF THE MODEL.

	Computation		Communication
	Total time	Time/iter.	Parameter/iter.
FedAvg	15.13 min	0.038 min	$2 * \Sigma$
SFL	15.62 min	0.039 min	Σ
Local-only	13.68 min	0.034 min	–
APFL	26.62 min	0.067 min	$2 * \beta_{\text{base}} * \Sigma$
FedPer	13.65 min	0.034 min	$2 * \beta_{\text{base}} * \Sigma$
FedProto	18.70 min	0.047 min	$2 * \beta_{\text{proto}} * \Sigma$
FedRep	13.65 min	0.034 min	$2 * \beta_{\text{rep}} * \Sigma$
FLOCO	18.95 min	0.047 min	$2 * \beta_{\text{endpoints}} * \Sigma$
LG-FedAvg	13.90 min	0.035 min	$2 * \beta_{\text{cls}} * \Sigma$
pFedFDA	13.90 min	0.035 min	$2 * (\beta_{\text{base}} + \mu_{\text{cls}} + \sigma_{\text{cls}}) * \Sigma$
SeqFedEDT	15.85 min	0.040 min	Σ

cost, we can observe that SeqFedEDT has a total computational time of 15.85 minutes (400 rounds) and a time per iteration of 0.040 minutes. This places SeqFedEDT in a competitive position when compared to other PFL methods like FedPer and FedRep, which have total times of 13.65 minutes and a time per iteration of 0.034 minutes. Our method only adds a small amount of communication time per iteration (0.006 minutes).

For communication cost, SeqFedEDT transmits Σ parameters per iteration, which has advantages over traditional PFL solutions. Other PFL methods also provide lower communication ($\beta_* \in (0, 1)$) costs compared to FedAvg. For example, APFL and FedPer only share the parameters of the base layers with the server. In contrast, LG-FedAvg only uploads the parameters in the classifier layer. FedRep learns a model representation after local training and exchanges this representation between clients and the server. FedProto designs a prototype-based structure and shares only the prototype parameters. Similarly, FLOCO also designs a multi-layer structure called endpoints and shares the parameters across those layers. pFedFDA increases the number of shared parameters by requiring the sharing of the base layer, along with the mean and variance of each parameter in the classifier layers during the training process. Leveraging its ring communication structure, SeqFedEDT transfers only half the parameters required by FedAvg, thereby offering a favorable trade-off between computation and communication costs over existing methods.

2) Computation Overhead: Fig. 7 presents the accuracy evolution with respect to total wall-clock training time on *CIFAR10* and *CIFAR100* under 400 epochs. The results indicate that SeqFedEDT incurs a notable increase in runtime when employing the Fisher metric, whereas PI and Grad. introduce only marginal overhead relative to other methods. Nevertheless, SeqFedEDT achieves faster accuracy gains within comparable time budgets, underscoring its efficiency advantage.

To further quantify this cost, Table VII reports the per-round runtime of computing metrics + decoupling (CMD) and its proportion of the total training time. Fisher consistently accounts for 70% across both simple (*MNIST*, *FMNIST*) and complex datasets (*SVHN*, *CIFAR100*), while PI and Grad. remain stable at 25–30%. On *MEDMNISTA/C*, the backbone differs: *LeNet5*,

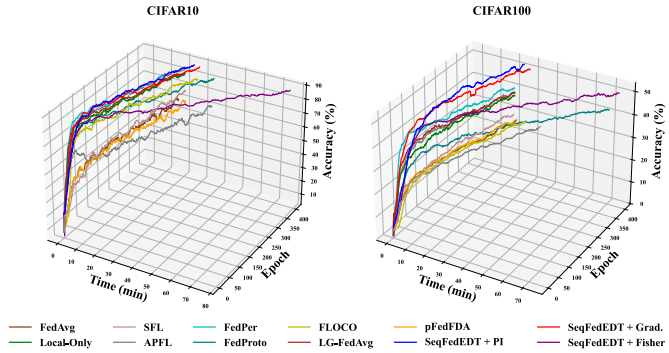


Fig. 7. Test accuracy as a function of wall-clock training time and epochs on CIFAR10 and CIFAR100.

TABLE VII
PER-ROUND TIME COST (MINUTES) AND PROPORTION FOR PI/Grad./Fisher METRICS COMPUTATION AND DECOUPLING IN SeqFedEDT

Dataset	PI		Grad.		Fisher	
	time	prop.	time	prop.	time	prop.
CIFAR100	0.019	0.291	0.021	0.295	0.133	0.721
CIFAR10	0.017	0.295	0.021	0.300	0.138	0.730
EMNIST	0.111	0.171	0.126	0.174	1.334	0.720
FMNIST	0.019	0.288	0.021	0.288	0.144	0.741
M.MNISTA	0.136	0.762	0.217	0.806	0.220	0.803
M.MNISTC	0.096	0.808	0.098	0.798	0.097	0.804
MNIST	0.020	0.288	0.021	0.288	0.146	0.744
SVHN	0.024	0.250	0.027	0.251	0.216	0.738

with substantially fewer parameters and lower-resolution inputs, reduces forward/backward costs and enlarges the relative share of CMD. Although Fisher is intrinsically more demanding due to per-sample gradient computations, the small parameterization of *LeNet5* yields tiny tensors, compressing per-sample autograd.grad calls. Consequently, Fisher's runtime is markedly reduced and approaches that of PI and Grad.

VIII. CONCLUSIONS AND FUTURE WORK

We proposed a novel SFL framework named SeqFedEDT that effectively addresses the challenges of SFL by leveraging model decoupling. Drawing on existing scoring methods in PFL and established concepts, we propose three different element-wise model decoupling approaches. Although PI has been validated in various fields, including large language models, its application within the context of SFL has largely been unexplored. Our work addresses this gap by introducing a novel quantile-based thresholding mechanism that differentiates between shared and personalized parameter subsets for SFL training. We conducted six primary experiments, including assessments of accuracy, convergence, scoring analysis, ablation studies, scalability and communication overhead analysis. The results demonstrate that SeqFedEDT consistently outperforms eight SOTA methods across various data heterogeneous scenarios and exhibits faster convergence compared to the original SFL.

Our work establishes a foundation for refined personalized parameter selection in SFL through scoring-based decoupling.

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This approach involves approximations using parameter differences, which may introduce small errors that accumulate across sequential client updates. To address this, future directions will focus on adaptive thresholding methods that respond to diverse data distributions, minimizing error propagation, and learnable parameter merging to balance personalization with generalization. We also plan to investigate the impact of SFL's path-dependent model delivery on parameter importance metrics, optimizing methods to mitigate errors in sequential updates. Meanwhile, the framework's peer-to-peer communication design shows potential in bandwidth-constrained settings [57], [58] such as remote healthcare and IoT [59], [60]. Parameter-level decoupling further supports robust personalization under high heterogeneity, making `SeqFedEDT` particularly promising for domains requiring personalized solutions.

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